

Secure-Tunnel: Post-Quantum Cryptographic Overlay for UAV C2 Links

Results and Analysis Section — v3.0 (2026-03-03)

Draft compilation for review

TABLE I: Algorithm Aliases and NIST Security Levels

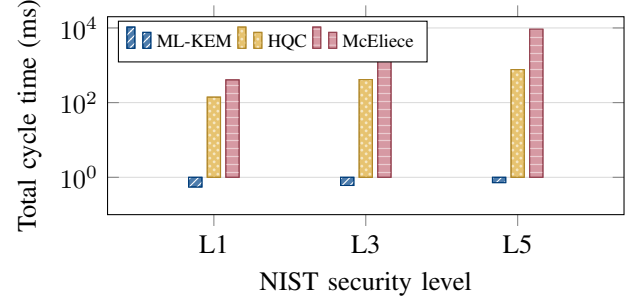
Alias	Full Name	Level	Type
ML512	ML-KEM-512	L1	KEM
ML768	ML-KEM-768	L3	KEM
ML1024	ML-KEM-1024	L5	KEM
HQ128	HQC-128	L1	KEM
HQ192	HQC-192	L3	KEM
HQ256	HQC-256	L5	KEM
MC348	Classic-McEliece-348864	L1	KEM
MC460	Classic-McEliece-460896	L3	KEM
MC8192	Classic-McEliece-8192128	L5	KEM
DSA44	ML-DSA-44	L1	SIG
DSA65	ML-DSA-65	L3	SIG
DSA87	ML-DSA-87	L5	SIG
F512	Falcon-512	L1	SIG
F1024	Falcon-1024	L5	SIG
SP128	SPHINCS ⁺ -SHA2-128s	L1	SIG
SP192	SPHINCS ⁺ -SHA2-192s	L3	SIG
SP256	SPHINCS ⁺ -SHA2-256s	L5	SIG
AESG	AES-256-GCM	—	AEAD
CH20	ChaCha20-Poly1305	—	AEAD
ASC	Ascon-128a	—	AEAD

I. RESULTS AND ANALYSIS

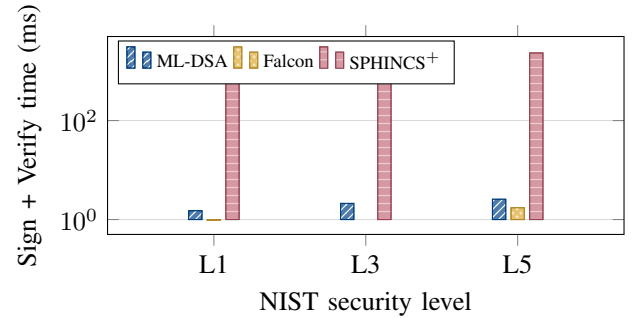
All measurements were collected on a Raspberry Pi 4 Model B Rev 1.5 (BCM2711, four Cortex-A72 cores at 1.8 GHz, 4 GB RAM) running Debian 12 (kernel 6.12.47) with the performance CPU governor. Timing used `time.perf_counter_ns()` with garbage collection disabled; power was sampled via an INA219 current sensor (0.1 Ω shunt, V_{BUS} gain correction 1.22) at up to 1 kHz. Idle power: $P_{idle} = 3.32$ W (30 s baseline). The Cortex-A72 lacks ARMv8 Crypto Extensions; AES-GCM therefore executes in software table-lookup mode, while ChaCha20-Poly1305 exploits 128-bit NEON SIMD. PQC primitives: liboqs-python 0.14.0. AEAD ciphers: OpenSSL 3.5.4 (AES-256-GCM, ChaCha20-Poly1305), ascon-c v1.2 opt64 via ctypes (Ascon-128a, KAT-verified). Table ?? defines the short aliases used throughout.

A. Primitive-Level Performance

Key Encapsulation. Fig. ?? and Table ?? compare nine KEM algorithms across three NIST security levels (100 iterations, 1 kHz INA219). Each cycle comprises keygen, encaps-



(a) KEM: keygen + encaps + decaps



(b) SIG: sign + verify (Falcon unavailable at L3)

Fig. 1: Primitive-level computational cost by NIST level (log scale). ML-KEM and ML-DSA/Falcon complete in < 3 ms at all levels; SPHINCS⁺ and McEliece are 2–4 orders of magnitude slower.

sulation, and decapsulation—all three are executed per rekey event. Fig. ?? presents both KEM and signature results using consistent log-scale grouping.

ML-KEM completes a full cycle under 0.72 ms through Level 5, drawing 3.46–3.48 W. HQC is 255–1 066 $\times$ slower, driven by decapsulation (73–393 ms). Classic McEliece exhibits extreme keygen asymmetry: MC8192 requires 9.02 s for keygen but only 1.89 ms for encapsulation, making it feasible only with pre-provisioned keys.

Digital Signatures. Fig. ?? and Table ?? evaluate eight signature schemes. Keygen is a one-time cost amortized over the key lifetime (0.4–282 ms); the per-handshake sign and

TABLE II: KEM Per-Operation Breakdown (100 iter, 1 kHz INA219)

Alias	Lvl	KeyGen (ms)	Encaps (ms)	Decaps (ms)	E_{tot} (μJ)	ρ
ML512	L1	0.240	0.152	0.158	1907	1.00×
HQ128	L1	22.33	44.68	73.39	535 253	255×
MC348	L1	351.0	0.466	55.98	1 737 186	741×
ML768	L3	0.237	0.182	0.185	2 112	1.00×
HQ192	L3	67.68	135.8	211.9	1 651 813	688×
MC460	L3	1 281	0.872	89.59	6 025 434	2 272×
ML1024	L5	0.273	0.216	0.231	2 507	1.00×
HQ256	L5	124.4	250.2	392.6	3 136 867	1 066×
MC8192	L5	9 021	1.894	210.1	40 583 491	12 823×

E_{tot} : total energy (keygen+encaps+decaps), INA219-measured. ρ : total cycle time relative to ML-KEM at same level.

TABLE III: Signature Runtime Cost: Sign + Verify (100 iter, 1 kHz INA219). KeyGen is a one-time cost and excluded from ρ .

Alias	Lvl	Sign (ms)	Verify (ms)	E_s (μJ)	E_v (μJ)	ρ
DSA44	L1	1.179	0.331	4 264	1 175	1.00×
F512	L1	0.778	0.192	2 773	683	0.64×
SP128	L1	1 468	1.582	6 008 719	5 690	972×
DSA65	L3	1.633	0.484	5 849	1 723	1.00×
SP192	L3	2 610	2.299	10 628 558	8 318	1 234×
DSA87	L5	1.847	0.723	6 612	2 564	1.00×
F1024	L5	1.452	0.280	5 175	995	0.67×
SP256	L5	2 321	3.292	9 516 164	11 738	904×

ρ : combined sign+verify time relative to ML-DSA at same level. Falcon $\rho < 1$: faster than ML-DSA. E_s/E_v : INA219-measured energy per operation.

verify operations determine rekey latency.

Falcon achieves the lowest combined sign+verify latency at L1 (0.970 ms) and L5 (1.732 ms), outperforming ML-DSA by 36% and 33% respectively. SPHINCS⁺ imposes prohibitive signing latency: SP128 consumes 1.47 s and 6.01 J per signature, approaching the rekey period at short intervals. Verification is fast (≤ 3.3 ms) for all algorithms, confirming the ground-station path is never the bottleneck.

Takeaway: Only ML-KEM paired with ML-DSA or Falcon enables sub-millisecond primitive costs at all NIST levels on resource-constrained ARM hardware.

B. Protocol-Level Handshake Performance

Each of the 24 level-matched KEM×SIG combinations was benchmarked end-to-end on the tunnel prototype (10 s per suite, baseline with no DDoS detector). T_{HS} is the measured wall-clock time from handshake initiation to session establishment, including protocol framing, HKDF key derivation, and interpreter overhead. T_{crypto} is the sum of in-suite KEM and SIG primitive times recorded during the same run.

Fig. ?? compares best-case handshake time per KEM family at each security level. Table ?? reports all 24 suites.

The default suite (ML768+DSA65) completes a full handshake in 12.71 ms at Level 3, of which only 2.71 ms (21%) is cryptographic computation; the remaining ~ 10 ms comprises

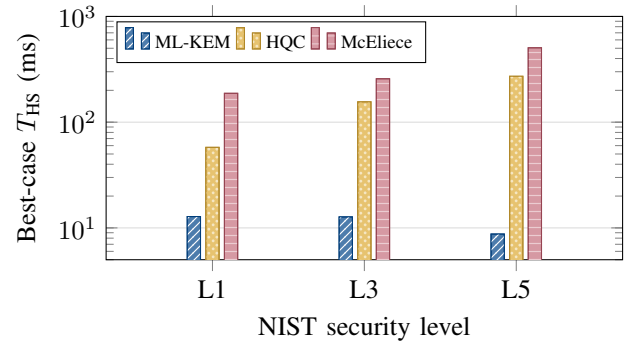


Fig. 2: Best-case measured handshake time per KEM family and NIST level (fastest SIG pairing, log scale). ML-KEM achieves < 13 ms at all levels; McEliece exceeds 500 ms at L5.

protocol framing, HKDF derivation, and interpreter overhead. ML1024+F1024 at Level 5 measures 8.75 ms—the fastest of all 24 suites—because Falcon-1024’s combined sign+verify (1.18 ms) undercuts ML-DSA-65 (1.94 ms).

Three structural findings emerge: (1) ML-KEM suites are the only family achieving sub-100 ms handshakes at every level ($\rho \leq 1.9\times$); (2) SPHINCS⁺ renders any suite non-viable regardless of KEM choice ($T_{\text{HS}} \geq 655$ ms); (3) McEliece at L5 exceeds 500 ms due to keygen asymmetry. The MC460+SP192 timeout at 48.2 s empirically justifies the scheduler’s constraint forbidding McEliece×SPHINCS⁺ pairs.

Takeaway: Protocol overhead dominates crypto cost for lattice-based suites; ML-KEM×ML-DSA/Falcon handshakes complete under 17 ms through NIST Level 5.

C. Data-Plane AEAD Performance

AEAD selection operates on the UDP data plane and does not affect handshake timing. Table ?? reports results from 10 000 iterations (500 warmup, two-pass methodology: timing with GC disabled, power with INA219 at ~ 100 Hz).

Timing is concentrated (σ_{enc} : AESG 0.38 μs , ASC 0.77 μs). CH20 exhibits $\sigma = 3.99 \mu\text{s}$ from rare scheduling jitter (P95=6.46, P99=6.69 μs ; median 6.30 μs).

Fig. ?? shows latency scaling from MAVLink telemetry (9 B) to bulk transfer (4096 B).

At 4096 B, CH20 is 3.93× faster than AESG due to NEON vectorization versus software AES table lookups. Below 64 B, setup cost dominates and the ciphers converge. Energy per bit at 256 B: CH20=21.8, AESG=32.2, ASC=46.4 nJ/bit. For a 50 Hz MAVLink stream (~ 256 B packets), per-second AEAD cost ranges from 2.23 mJ (CH20) to 4.75 mJ (ASC)—negligible versus the 3.32 W board draw.

Takeaway: ChaCha20-Poly1305 delivers 32% lower energy per bit than AES-256-GCM on Cortex-A72 without hardware AES, making it the optimal data-plane cipher for this platform.

D. Adversarial Interaction Impact

Concurrent DDoS detection competes with the cryptographic pipeline for CPU and thermal headroom. All 72

TABLE IV: Measured End-to-End Suite Handshake Performance — All 24 Level-Matched KEM×SIG Combinations (Baseline, No DDoS Detector)

KEM	SIG	T_{HS} (ms)	T_{crypto} (ms)	$\hat{E}_{\text{HS}}$ (mJ)	Crypto (%)	ρ	Viable?
<i>NIST Level 1</i> (3 KEMs × 3 SIGs = 9 suites)							
ML512	F512	12.80	5.69	44.3	44	1.0×	✓
ML512	DSA44	14.34	2.45	49.6	17	1.1×	✓
HQ128	F512	57.80	54.30	200	94	4.5×	✓
HQ128	DSA44	61.43	56.50	213	92	4.8×	✓
MC348	DSA44	187.5	132.3	649	71	14.6×	✓*
MC348	F512	206.1	161.4	713	78	16.1×	✓*
ML512	SP128	655.1	644.8	2 268	98	51.2×	×
HQ128	SP128	722.8	718.1	2 502	99	56.5×	×
MC348	SP128	1 080	1 045	3 739	97	84.3×	×
<i>NIST Level 3</i> (3 KEMs × 2 SIGs = 6 suites)							
ML768	DSA65	12.71	2.71	44.3	21	1.0×	✓
HQ192	DSA65	155.8	159.9	543	100 [‡]	12.3×	✓*
MC460	DSA65	257.4	185.4	897	72	20.2×	✓*
ML768	SP192	1 250	1 233	4 362	99	98.3×	×
HQ192	SP192	1 537	1 533	5 364	100	120.9×	×
MC460	SP192	48 186 [†]	—	—	—	—	×
<i>NIST Level 5</i> (3 KEMs × 3 SIGs = 9 suites)							
ML1024	F1024	8.75	1.82	30.5	21	1.0×	✓
ML1024	DSA87	16.78	2.25	58.5	13	1.9×	✓
HQ256	F1024	271.7	283.7	947	100 [‡]	31.1×	✓*
HQ256	DSA87	276.1	288.9	963	100 [‡]	31.6×	✓*
MC8192	DSA87	505.3	367.3	1 763	73	57.7×	×
MC8192	F1024	678.4	527.9	2 367	78	77.5×	×
ML1024	SP256	1 110	1 092	3 873	98	126.9×	×
HQ256	SP256	1 470	1 494	5 128	100 [‡]	168.0×	×
MC8192	SP256	1 651	1 525	5 759	92	188.7×	×

T_{HS} : measured end-to-end handshake time. T_{crypto} : sum of in-suite KEM and SIG primitive times. $\hat{E}_{\text{HS}}$: estimated energy ($P_{\text{avg}} \times T_{\text{HS}}$, using per-primitive power from Tables ??–??). ρ : relative to fastest suite at same NIST level. Viable: ✓ $T_{\text{HS}} < 100$ ms; ✓* extended rekey; × $T_{\text{HS}} > 500$ ms or timeout. [†]Timeout at 48.2 s; crypto fields unavailable. [‡] $T_{\text{crypto}} > T_{\text{HS}}$ (measurement jitter); capped at 100%.

TABLE V: AEAD Performance at 256-Byte Payload (10 000 iter, RPi4). Independent verification run; native C Ascon-128a backend (ascon-c v1.2 opt64).

Cipher	Enc (μ s)	Dec (μ s)	P_{enc} (W)	P_{dec} (W)	E_{pkt} (μ J)	E_{bit} (nJ/bit)
AESG	9.25	9.09	3.63	3.57	66.0	32.2
CH20	6.47	6.05	3.54	3.58	44.6	21.8
ASC	12.72	12.85	3.84	3.59	95.0	46.4

$E_{\text{pkt}} = P_{\text{avg}} \times (t_{\text{enc}} + t_{\text{dec}})$. $E_{\text{bit}} = E_{\text{pkt}}/2048$. Board-level power ($P_{\text{idle}} = 3.320$ W not subtracted). ΔP above idle: AESG 0.31, CH20 0.22, ASC 0.52 W.

level-matched suites were benchmarked under baseline (no detector), XGBoost, and TST scenarios (10 s per suite per scenario).¹

Across 72 suites, XGBoost adds mean +2.5% / median +0.05% handshake overhead (max +137.2%), while TST adds mean +71.8% / median +32.6% (max +405.1%). For the default suite (ML768+DSA65), T_{HS} rises from 13.01 ms baseline to 13.46 ms with XGBoost (+3.5%) and 37.90 ms with TST (+191.3%). TST’s 93% CPU utilization starves the

¹Lightweight detectors use 2 features (Mavlink_Count, Total_length); TST uses seq_len=1, functionally a 3-layer MLP with 1.6 M parameters.

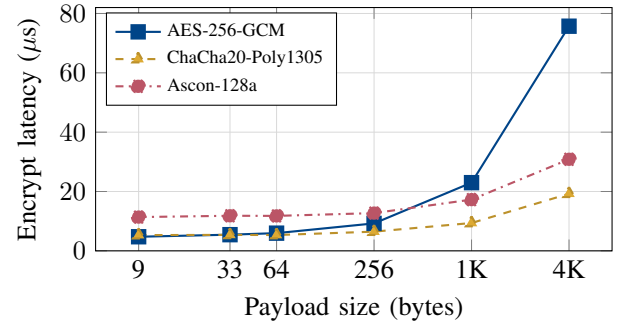


Fig. 3: AEAD encrypt latency versus payload size. ChaCha20 (NEON SIMD) scales at 3.4 μ s/KiB; AES-GCM at 17.7 μ s/KiB (software table-lookup). Below 64 B, setup cost dominates and ciphers converge.

handshake thread.

Table ?? and Fig. ?? compare four CIC-IoT-2023 detectors on inference cost.

XGBoost achieves the highest accuracy (94.55%) at 7.42 ms mean latency and 31.7 mJ per inference. RandomForest consumes 10× more energy (291.4 mJ) due to its 200-tree ensemble. TST delivers the lowest accuracy (90.27%) at the highest power cost (+1 973 mW, 93% CPU), with a P99 tail

TABLE VI: DDoS Detector Inference on RPi4 (CIC-IoT-2023, 3.4 M samples)

Model	Acc. (%)	F1	t_{inf} (ms)	P99 (ms)	ΔP (mW)	E_{inf} (mJ)
LightGBM	93.47	0.937	6.72	8.59	+999	29.0
XGBoost	94.55	0.947	7.42	9.52	+951	31.7
RF	93.35	0.935	74.68	102.63	+594	291.4
TST	90.27	0.80	10.72	26.12	+1 973	56.7

Accuracy/F1 from model evaluation; inference metrics measured on RPi4 (15 s per model, idle baseline 3.309 W). XGB/LGBM/RF: 54 features, 15 classes. TST: 46 features, 6 classes.

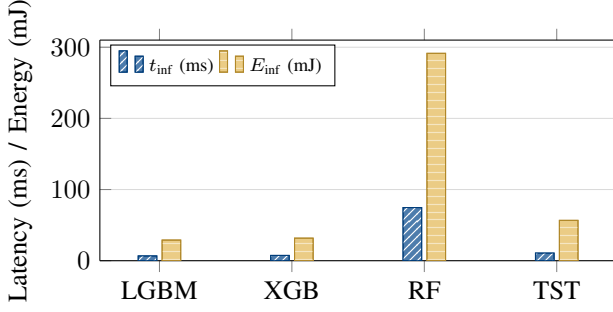


Fig. 4: DDoS detector inference cost on RPi4. XGBoost and LightGBM are Pareto-optimal (< 8 ms latency, < 32 mJ energy). RandomForest is $10\times$ more costly on both axes.

of 26.12 ms.

Takeaway: Tree-based models (XGBoost, LightGBM) are Pareto-optimal for embedded PQC-DDoS co-deployment, adding $< 4\%$ handshake overhead at the highest detection accuracy.

E. Scheduler Implications

The Multi-Dimensional Energy-Aware Scheduler (MDEAS) manages three decision axes at 1 Hz: AEAD cipher selection, security-level promotion/demotion (triggering PQC rekey), and DDoS detector activation. Each level transition requires a full handshake. The rekey overhead fraction is:

$$\Phi(R) = \frac{T_{\text{HS}}}{R + T_{\text{HS}}} \quad (1)$$

where R is the rekey interval. For the default suite ($T_{\text{HS}} = 12.71$ ms) at the minimum 30 s interval: $\Phi = 4.2 \times 10^{-4}$ ($< 0.05\%$). Even ML768+SP192 ($T_{\text{HS}} = 1.25$ s) at 30 s yields $\Phi = 0.040$ (4.0%).

Table ?? lists the per-detector resource profiles used for proactive thermal management.

Cross-axis constraints prevent pathological configurations: Ascon-128a is forbidden under TST (per-packet cost plus 93% CPU exceeds the real-time budget); McEliece $\times$ SPHINCS⁺ pairs are forbidden (T_{HS} exceeds the 45 s rekey timeout); detector activation is blocked above the per-detector T_{max} . These constraints reduce 648 theoretical configurations (216 suites $\times$ 3 detector states) to ~ 500 viable operating points.

TABLE VII: MDEAS Detector Thermal and CPU Overhead

Detector	ΔP (W)	ΔT ($^{\circ}\text{C}$)	ΔCPU (pp)	T_{max} ($^{\circ}\text{C}$)
None	0.00	0.0	0	80.0
XGBoost	0.95	4.8	35	75.0
TST	1.97	10.7	91	69.0

T_{max} : temperature ceiling before activation is blocked (80°C throttle $-\Delta T$). ΔCPU in percentage points.

Takeaway: Rekey overhead is negligible ($< 0.05\%$) for ML-KEM suites. The MDEAS navigates the security–energy–thermal Pareto surface via 1 Hz telemetry while guaranteeing the configured minimum security level.